

Database Design

Overview

This document serves as the single source of truth for the database design of the Animal Adoption Management System (AAMS). It covers all stages of the design process, from initial entity and relationship identification through to cardinality mapping, table definitions with column names and data types, and the final entity-relationship diagram. All design decisions are grounded in the functional requirements defined for the six user roles: Admin, Shelter Staff, Adopter, Veterinarian, Volunteer, and Donor.

1. Entities & Relationships

This section maps each functional requirement to its corresponding entities and relationships. Relationships are expressed in the format ENTITY_A — verb — ENTITY_B, where each entry represents a structural dependency that will translate into a table, foreign key, or junction table in the final schema. Query-only operations (such as search and filter) are excluded, as they do not produce new schema elements. Relationships are grouped by user role to reflect the access and ownership model of the system.

1.1 Adopter

No	Relationship
01	ADOPTER - has - GOVERNMENT_ID
02	ADOPTER - has - FAVORITES
03	ADOPTER - submits - ADOPTION_APPLICATION
04	ADOPTER - receives - NOTIFICATIONS
05	ADOPTER - cancels - VISITS

1.2 Shelter Staff

No	Relationship
01	STAFF - has - GOVERNMENT_ID
02	STAFF - updates - PET
03	STAFF - belongs_to - SHELTER
04	STAFF - manages - APPLICATIONS
05	STAFF - updates - VOLUNTEERS
06	STAFF - updates - VOLUNTEER_TASKS

No	Relationship
07	STAFF - updates - SHELTER_EVENTS
08	STAFF - updates - DONATIONS
09	STAFF - manages - VISITS

1.3 Pet

No	Relationship
01	PET - belongs_to - BREED
02	BREED - belongs_to - SPECIES
03	PET - has - HEALTH_RECORD
04	PET - has - VACCINATION_RECORD
05	PET - has - TRANSFER_HISTORY

1.4 Veterinarian

No	Relationship
01	VET - has - GOVERNMENT_ID
02	VET - manages - APPOINTMENTS
03	VET - updates - VACCINATION_RECORD
04	VET - updates - HEALTH_RECORD

1.5 Volunteer

No	Relationship
01	VOLUNTEER - HAS - GOVERNMENT_ID
01	VOLUNTEER - submits - VOLUNTEER_APPLICATION
02	VOLUNTEER - has - TASKS
03	VOLUNTEER - manages - SHELTER_EVENTS

1.6 DONOR

No	Requirement
01	DONOR - makes - DONATIONS

1.7 Admin

No	Requirement
01	ADMIN - has - GOVERNMENT_ID
02	ADMIN - creates - SHELTER
03	ADMIN - assigns - STAFF

2. Cardinality

This section defines the cardinality of each relationship identified in Section 1. Cardinality describes how many instances of one entity can be associated with instances of another, and is expressed as 1:1, 1:M, or M:M. Many-to-many relationships are resolved into junction tables, which are listed alongside the relationship. These mappings directly inform the placement of foreign keys and the creation of junction tables in the final schema.

No	Relationship	Cardinality	Junction
01	ADOPTER - has - GOVERNMENT_ID	1:1	-
02	ADOPTER - has - FAVORITES	M:M	FAVORITES (adopterID, petID)
03	ADOPTER - submits - ADOPTION_APPLICATION	1:M	-
04	ADOPTER - cancels - VISITS	1:M	-
05	ADOPTER - receives - NOTIFICATIONS	1:M	-
06	STAFF - has - GOVERNMENT_ID	1:1	-
07	STAFF - updates - PET	1:M	-
08	STAFF - belongs_to - SHELTER	M:1	-
09	STAFF - manages - ADOPTION_APPLICATIONS	1:M	-
10	STAFF - updates - VOLUNTEER	1:M	-
11	STAFF - assigns - TASKS	1:M	-
12	VOLUNTEER - handles - TASKS	M:M	VOLUNTEER_TASKS (taskID, volunteerID)
13	STAFF - updates - EVENTS	1:M	-
14	STAFF - creates - DONOR	1:M	-
15	STAFF - manages - VISITS	1:M	-
16	PET - belongs_to - BREED	M:1	-
17	BREED - belongs_to - SPECIES	M:1	-
18	PET - has - HEALTH_RECORD	1:M	-
19	PET - has - VACCINATION_RECORD	1:M	-

No	Relationship	Cardinality	Junction
20	PET - has - TRANSFER_HISTORY	1:M	-
21	VET - manages - APPOINTMENTS	1:M	-
22	VET - updates - VACCINATION_RECORD	1:M	-
23	VET - updates - HEALTH_RECORD	1:M	-
24	VOLUNTEER - submits - VOLUNTEER_APPLICATION	1:1	-
25	VOLUNTEER - has - TASKS	1:M	-
26	VOLUNTEER - manages - SHELTER_EVENTS	M:M	VOLUNTEER_EVENT (eventID, volunteerID)
27	DONOR - makes - DONATIONS	1:M	-
28	ADMIN - creates - SHELTER	1:M	-
29	ADMIN - assigns - STAFF	1:M	-
30	APPOINTMENT - records - VACCINATION_RECORD	1:M	-
31	STAFF - assists - APPOINTMENTS	1:M	-
32	VOLUNTEER - assists - APPOINTMENTS	1:M	-

3. Data Tables

This section defines the complete schema for each of the tables in the system. For every table, all columns are listed with their data type, constraints, and a brief description of the field's purpose. Primary keys and foreign keys are explicitly marked. These definitions serve as the direct input for the Prisma schema and database migrations.

3.1 USERS

Central authentication table shared by all six role tables (Admin, Staff, Veterinarian, Adopter, Volunteer, Donor) via userID as a shared primary/foreign key.

Column Name	Data Type	Constraints	Description
userID	INT	PK, AUTO_INCREMENT	Unique identifier for the user, shared across all role tables
userEmail	VARCHAR (45)	NOT NULL, UNIQUE	Login email address
userPassword	VARCHAR (255)	NOT NULL	Bcrypt hashed password
refreshToken	TEXT	NULLABLE	Hashed refresh token for the two-token auth flow

Column Name	Data Type	Constraints	Description
role	VARCHAR (20)	NOT NULL ENUM ('Adopter', 'Volunteer', 'Donor', 'Admin', 'Staff', 'Veterinarian')	Designates which role table this user's profile row lives in
emailVerified	BOOLEAN	DEFAULT FALSE	Email verification status
lastLoginAt	DATETIME	NULLABLE	Most recent successful login (all roles)

3.2 ADMIN

Stores account and identity information for system administrators who manage shelter branches and staff assignments.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the admin, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
adminName	VARCHAR (45)	NOT NULL	Name of the admin
createdAt	DATETIME	NOT NULL	Account creation timestamp
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated')	Account status
adminPhone	VARCHAR (20)	NULLABLE	Admin contact number
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the admin has finished the onboarding wizard

Column Name	Data Type	Constraints	Description
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
adminDOB	DATE	NULLABLE	Date of birth of admin
adminSex	CHAR (1)	NULLABLE	Gender of admin
statusChangedBy ID	INT	FK → ADMIN (userID) NULLABLE ON DELETE SET NULL	The admin who last changed this admin's accountStatus (self-referencing)
statusChangedAt	DATETIME	NULLABLE	Timestamp of the last accountStatus change, paired with statusChangedByID

3.3 SHELTER

Stores details for each shelter branch in the multi-shelter network including location, capacity and status.

Column Name	Data Type	Constraints	Description
shelterID	INT	PK, AUTO_INCREMENT	Unique identifier for the shelter
shelterName	VARCHAR (45)	NOT NULL	Name of the shelter
shelterAddress	VARCHAR (45)	NOT NULL	Shelter location
shelterPhone	VARCHAR (20)	NOT NULL	Shelter phone number
shelterEmail	VARCHAR (45)	NOT NULL	Shelter email address
shelterLocation	GEOGRAPHY (POINT)	NULLABLE	Location of the shelter (PostGIS)
shelterZIP	INT	NOT NULL	Zip code of shelter
shelterSize	INT	NOT NULL	Capacity of shelter
shelterStatus	VARCHAR (10)	DEFAULT 'Open' ENUM ('Open', 'Full', 'Closed')	Shelter Status
managerStaffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the shelter manager

3.4 STAFF

Stores account, contact and employment information for shelter staff members.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the staff, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
staffName	VARCHAR (45)	NOT NULL	Full name of the staff
staffPhone	VARCHAR (20)	NULLABLE	Staff contact number
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the staff has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	The shelter branch this staff works at
staffDOB	DATE	NULLABLE	Date of birth of staff
staffSex	CHAR (1)	NULLABLE	Gender of staff
staffDOJ	DATETIME	NULLABLE	Staff joining date
staffDOS	DATETIME	NULLABLE	Staff date of separation
staffDesignation	VARCHAR (45)	NULLABLE ENUM ('Manager', 'Senior', 'Associate')	Staff designation
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated')	Account Status

3.5 VETERINARIAN

Stores account and contact information for vets who manage animal health appointments and records.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the vet, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
vetName	VARCHAR (45)	NOT NULL	Name of the vet
addressLine1	VARCHAR (100)	DEFAULT ""	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ""	City
state	VARCHAR (45)	DEFAULT ""	State
zip	VARCHAR (10)	DEFAULT ""	Postal code
country	VARCHAR (45)	DEFAULT ""	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the vet has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
vetPhone	VARCHAR (20)	NULLABLE	Vet contact number
vetDOB	DATE	NULLABLE	Date of Birth of vet
vetSex	CHAR (1)	NULLABLE	Gender of vet
createdAt	DATETIME	NOT NULL	Account creation timestamp
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	Unique identifier for the shelter vet associated with

Column Name	Data Type	Constraints	Description
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Deactivated')	Account Status

3.6 SPECIES

Stores the list of animal species supported by the system (e.g. Dog, Cat, Rabbit).

Column Name	Data Type	Constraints	Description
speciesID	INT	PK, AUTO_INCREMENT	Unique identifier for the species
speciesName	VARCHAR (45)	NOT NULL	Name of the species

3.7 BREED

Stores animal breeds and their association to a parent species.

Column Name	Data Type	Constraints	Description
breedID	INT	PK, AUTO_INCREMENT	Unique identifier for the breed
speciesID	INT	FK → SPECIES (speciesID)	Unique identifier for the species the breed belongs to
breedName	VARCHAR (45)	NOT NULL	Name of the breed

3.8 PET

Stores profile, intake, medical and compatibility information for all animals in the shelter network.

Column Name	Data Type	Constraints	Description
petID	INT	PK, AUTO_INCREMENT	Unique identifier for the pet.
petCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "PE000002") GENERATED ALWAYS STORED from petID, read-only
petName	VARCHAR (45)	NOT NULL	Name of the pet.
breedID	INT	FK → BREED (breedID)	Pet Breed
petDOB	DATE	NOT NULL	Date of birth of the pet

Column Name	Data Type	Constraints	Description
petWeight	FLOAT	NOT NULL	Pet weight
petHeight	FLOAT	NOT NULL	Pet height
petBGroup	VARCHAR (5)	NOT NULL	Pet blood group
petColor	VARCHAR (45)	NOT NULL	Pet color
petSize	VARCHAR (10)	NULLABLE ENUM ('Small', 'Medium', 'Large')	Size of pet
petPhoto	VARCHAR (300)	NOT NULL	Primary profile photo URL for the Pet
petSex	CHAR (1)	NOT NULL	Pet gender.
petDesc	VARCHAR (500)	NULLABLE	Free-text description of the pet's personality, temperament and behaviour.
microchipID	VARCHAR (45)	NULLABLE	Industry-standard microchip identifier implanted on the pet. Nullable if not yet microchipped.
intakeDate	DATE	NOT NULL	Most recent intake date
intakeType	VARCHAR (45)	NULLABLE ENUM ('stray', 'surrendered', 'transferred')	Most recent intake type
adoptionStatus	VARCHAR (45)	NULLABLE ENUM ('incoming', 'available', 'adopted', 'fostered', 'transferred', 'deceased')	Availability
shelterID	INT	FK → SHELTER (shelterID)	The shelter branch where the pet is currently housed.
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff member managing this pet's profile
compatibleWithChildren	BOOLEAN	DEFAULT FALSE	Compatibility with children
compatibleWithPets	BOOLEAN	DEFAULT FALSE	Compatibility with other pets

Column Name	Data Type	Constraints	Description
specialNeeds	BOOLEAN	DEFAULT FALSE	Special needs requirements
featuredFlag	BOOLEAN	DEFAULT FALSE	Marks the pet as featured on the Home page

3.9 PET_PHOTO

Stores one or more photos per pet. Replaces the single petPhoto field with a one-to-many gallery; PET.petPhoto remains as the primary/profile photo URL, with additional photos stored here.

Column Name	Data Type	Constraints	Description
photoID	INT	PK, AUTO_INCREMENT	Unique identifier for the photo
petID	INT	FK → PET (petID)	Unique identifier for the pet this photo belongs to
photoURL	VARCHAR (300)	NOT NULL	Photo URL (Supabase Storage)
uploadedAt	DATETIME	NOT NULL DEFAULT now()	Upload timestamp

3.10 Adopter

Stores personal, household, lifestyle and account information for users who register to adopt or foster a pet.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the adopter, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
adopterName	VARCHAR (45)	NOT NULL	Full name of the adopter
adopterDOB	DATE	NULLABLE	Date of birth of adopter
adopterSex	CHAR (1)	NULLABLE	Gender of adopter
createdAt	DATETIME	NOT NULL	Account creation timestamp
adopterRiskFlag	BOOLEAN	DEFAULT FALSE	Flag set if AI feature detects suspicious adoption history

Column Name	Data Type	Constraints	Description
preQualifyFlag	BOOLEAN	DEFAULT FALSE	Flag set if adopter is pre-qualified based on experience and history, used to fast-track applications.
adopterPhone	VARCHAR (20)	NULLABLE	Adopter contact number
housingType	VARCHAR (15)	NULLABLE ENUM ('Apartment', 'House', 'Other')	House type
ownsOrRents	VARCHAR (10)	NULLABLE ENUM ('Owns', 'Rents')	House ownership status
landlordContact	VARCHAR (20)	NULLABLE	Landlord contact
householdSize	INT	NULLABLE	Household size
numChildren	INT	NULLABLE	Number of children at home
employmentStatus	VARCHAR (15)	NULLABLE ENUM ('Unemployed', 'Student', 'Self_employed', 'Employed')	Employment status of the adopter
activityLevel	VARCHAR (10)	NULLABLE ENUM ('Sedentary', 'Medium', 'Active')	Level of activity in lifestyle of adopter
yardAvailable	BOOLEAN	DEFAULT FALSE	Yard availability
petExperience	VARCHAR (10)	NULLABLE ENUM ('No', 'Little', 'Some', 'Very')	Amount of experience with pets
currentPets	INT	DEFAULT 0	Number of pets currently owned by the adopter at time of registration.
preferredBreedID	INT	FK → BREED (breedID) NULLABLE	Preferred breed for adoption
preferredAgeRange	VARCHAR (15)	NULLABLE ENUM ('Young', 'Adult', 'Old')	Preferred pet age for adoption
preferredSize	VARCHAR (10)	NULLABLE ENUM ('Small', 'Medium', 'Large')	Preferred pet size for adoption
openToSpecialNeeds	BOOLEAN	DEFAULT FALSE	Open to adopting special needs pets

Column Name	Data Type	Constraints	Description
adopterType	VARCHAR (10)	NULLABLE ENUM ('Fosterer', 'Owner')	Type of adopter
stripeCustomerID	VARCHAR (255)	NULLABLE	Stripe payment ID
accountStatus	VARCHAR (45)	DEFAULT 'Active' ENUM ('Active', 'Banned', 'Deactivated')	Account status
addressLine1	VARCHAR (100)	DEFAULT ''	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ''	City
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the adopter has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)

3.11 FAVORITES

Junction table linking adopters to pets they have saved to their favorites list.

Column Name	Data Type	Constraints	Description
adopterID	INT	PK, FK → ADOPTER (userID)	Unique identifier for the adopter
petID	INT	PK, FK → PET (petID)	Unique identifier for the pet

3.12 ADOPTION_APPLICATION

Stores adoption applications submitted by adopters for specific pets, including status and assigned staff.

Column Name	Data Type	Constraints	Description
applicationID	INT	PK, AUTO_INCREMENT	Unique identifier for the application.
petID	INT	FK → PET (petID)	Unique identifier for the pet

Column Name	Data Type	Constraints	Description
adopterID	INT	FK → ADOPTER (userID)	Unique identifier for the adopter
createdAt	DATETIME	NOT NULL	Application creation timestamp
staffID	INT	FK → STAFF (userID) NULLABLE	Staff ID of Staff handling the application
shelterID	INT	FK → SHELTER (shelterID)	Shelter ID of Shelter application is associated with.
applicationStatus	VARCHAR (20)	DEFAULT 'Pending' ENUM ('Pending', 'Accepted', 'Rejected', 'Withdrawn')	Application status
applicationCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "APP-00048") — Postgres GENERATED ALWAYS STORED column computed from applicationID, read-only
applicationType	VARCHAR (10)	DEFAULT 'Adopt', ENUM ('Adopt', 'Foster')	Whether this is an adopt or foster application
shelterMessage	VARCHAR (500)	NULLABLE	Adopter's message to the shelter, submitted with the application
staffRemark	VARCHAR (300)	NULLABLE	Staff-authored closing note shown to the adopter once resolved
stripeCheckoutSessionID	VARCHAR (255)	NOT NULL	The Stripe Checkout Session that paid the application fee
stripePaymentIntentID	VARCHAR (255)	NULLABLE	The associated Stripe PaymentIntent
paymentStatus	VARCHAR (10)	DEFAULT 'Paid', ENUM ('Paid')	Payment status of the processing fee
amountPaid	DECIMAL (6,2)	DEFAULT 15.00	Application processing fee actually charged

3.13 VISITS

Stores shelter visit and virtual meet-and-greet appointments scheduled by adopters.

Column Name	Data Type	Constraints	Description
visitID	INT	PK, AUTO_INCREMENT	Unique identifier for the visit
adopterID	INT	FK → ADOPTER (userID)	Unique identifier for the adopter

Column Name	Data Type	Constraints	Description
petID	INT	FK → PET (petID) NULLABLE	Unique identifier for pet associated with the visit. Can be NULL
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for Staff assisting the visit. Can be NULL
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for shelter associated with the visit.
visitTime	DATETIME	NOT NULL	Visit date and time
remarks	VARCHAR (300)	NULLABLE	Any remarks related to the visit
visitStatus	VARCHAR (45)	NULLABLE ENUM ('Confirmed', 'Cancelled', 'Completed')	Visit status — null until a staff member confirms it

3.14 NOTIFICATIONS

Stores system-generated automated reminders sent to users for vaccinations, appointments and tasks.

Column Name	Data Type	Constraints	Description
notificationID	INT	PK, AUTO_INCREMENT	Unique identifier for the notification
recipientID	INT	NOT NULL	Unique identifier for the recipient
notificationType	VARCHAR (20)	ENUM ('vaccination', 'appointment', 'task')	Type of notification
notificationStatus	VARCHAR (10)	ENUM ('Pending', 'Sent', 'Failed')	Status of notification delivery
recipientType	VARCHAR (20)	ENUM ('adopter', 'staff', 'volunteer', 'veterinarian')	Recipient user type
sendTime	DATETIME	NOT NULL	Send date and time
textBody	VARCHAR (300)	NOT NULL	Notification text content

3.15 HEALTH_RECORD

Stores medical history entries for each pet including diagnosis, treatment and vet notes.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
createdAt	DATETIME	NOT NULL	Creation date
lastUpdated	DATETIME	NOT NULL	Last update date
vetID	INT	FK → VETERINARIAN (userID) NULLABLE	Vet associated with record
recordDesc	VARCHAR (500)	NOT NULL	Description of the medical visit, diagnosis, treatment, or special needs noted by the vet.

3.16 VACCINE

Column Name	Data Type	Constraints	Description
vaccineID	INT	PK, AUTO_INCREMENT	Unique identifier for the vaccine
vaccineName	VARCHAR (45)	NOT NULL	Name of the vaccine
manufacturer	VARCHAR (45)		Manufacturer name
vaccineDesc	VARCHAR (500)		Description of the vaccine

3.17 VACCINATION_RECORD

Stores vaccination history for each pet including vaccine details, dates and administering vet.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
vaccineID	INT	FK → VACCINE (vaccineID)	Unique identifier for the vaccine administered
administeredDate	DATETIME	NOT NULL	Administered date
dueDate	DATETIME	NOT NULL	Due date for administering vaccine

Column Name	Data Type	Constraints	Description
administeredBy	INT	FK → VETERINARIAN (userID) NULLABLE	Unique identifier for the vet administered the vaccine
administeredAt	INT	FK → SHELTER (shelterID) NULLABLE	Unique identifier for the shelter where vaccine was administered
appointmentID	INT	FK → APPOINTMENTS (appointmentID) NULLABLE ON DELETE SET NULL	The appointment the dose was given at; null for doses recorded outside an appointment.

3.18 TRANSFER_HISTORY

Stores the history of inter-shelter pet transfers including the shelters and staff involved.

Column Name	Data Type	Constraints	Description
recordID	INT	PK, AUTO_INCREMENT	Unique identifier for the record
petID	INT	FK → PET (petID)	Unique identifier for the pet the record belongs to
transferStatus	VARCHAR (20)	DEFAULT 'In_Progress' ENUM ('In_Progress', 'Completed', 'Rejected', 'Cancelled')	Transfer Status
transferDate	DATETIME	NOT NULL	Date of transfer
fromShelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the current shelter the pet is at
toShelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the new shelter the pet moving to
fromShelterStaff	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling transfer at current shelter
toShelterStaff	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling transfer at new shelter
transferReason	VARCHAR (300)	NOT NULL	Reason for transfer

3.19 APPOINTMENTS

Stores veterinary health appointments for pets including the attending vet and associated shelter.

Column Name	Data Type	Constraints	Description
appointmentID	INT	PK, AUTO_INCREMENT	Unique identifier for the appointmentID
appointmentCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "APT-00123") GENERATED ALWAYS STORED from appointmentID, read-only
petID	INT	FK → PET (petID)	Unique identifier for the pet associated with the appointment
vetID	INT	FK → VETERINARIAN (userID)	Unique identifier for the vet attending the appointment
staffID	INT	FK → STAFF (userID) NULLABLE	Staff member assigned
volunteerID	INT	FK → VOLUNTEER (userID) NULLABLE	Volunteer assigned
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter the appointment is at
appointmentDate	DATETIME	NOT NULL	Appointment date and time
appointmentReason	VARCHAR (300)	NOT NULL	Reason for appointment
appointmentStatus	VARCHAR (10)	DEFAULT 'Scheduled' ENUM ('Scheduled', 'Completed', 'Cancelled')	Appointment status

3.20 VOLUNTEER_APPLICATION

Stores applications submitted by prospective volunteers before their account is created.

Column Name	Data Type	Constraints	Description
applicationID	INT	PK, AUTO_INCREMENT	Unique identifier for the application
volunteerID	INT	FK → VOLUNTEER (userID) NULLABLE	Volunteer ID created only if application is approved
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff handling the application
createdAt	DATETIME	NOT NULL	Application creation date and time.
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for shelter

Column Name	Data Type	Constraints	Description
			volunteer is applying to
applicationStatus	VARCHAR (45)	DEFAULT 'Created' ENUM ('Created', 'Accepted', 'Rejected')	Application status
applicationRemark	VARCHAR (100)	NULLABLE	Any remarks

3.21 VOLUNTEER

Stores account, contact and availability information for approved volunteers.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the volunteer, shared with USERS
volunteerCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "VOL-00005") GENERATED ALWAYS STORED from userID, read-only
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
volunteerName	VARCHAR (45)	NOT NULL	Name of the volunteer
addressLine1	VARCHAR (100)	DEFAULT ""	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ""	City
state	VARCHAR (45)	DEFAULT ""	State
zip	VARCHAR (10)	DEFAULT ""	Postal code
country	VARCHAR (45)	DEFAULT ""	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the volunteer has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step

Column Name	Data Type	Constraints	Description
			reached (only ever advances)
volunteerPhone	VARCHAR (20)	NULLABLE	Volunteer contact number
volunteerDOB	DATE	NULLABLE	Date of Birth of volunteer
volunteerSex	CHAR (1)	NULLABLE	Gender of volunteer
volunteerSchedule	VARCHAR (100)	NULLABLE	Schedule of volunteer
shelterID	INT	FK → SHELTER (shelterID) NULLABLE	Unique identifier for shelter volunteer works at
createdAt	DATETIME	NOT NULL	Account creation timestamp
accountStatus	VARCHAR (45)	DEFAULT 'Pending' ENUM ('Pending', 'Active', 'Banned', 'Deactivated')	Account status — Pending until a staff member approves

3.22 GovernmentID

Stores government identity documents for adopters, staff, volunteers, vets and admins for verification purposes.

Column Name	Data Type	Constraints	Description
governmentIDID	INT	PK, AUTO_INCREMENT	Unique identifier for the record.
userID	INT	NOT NULL	The ID of the adopter, staff, volunteer, veterinarian or admin
userType	VARCHAR (45)	NOT NULL ENUM ('Adopter', 'Staff', 'Volunteer', 'Veterinarian', 'Admin')	Designation
idType	VARCHAR (45)	NOT NULL	Type of ID document
idNumber	VARCHAR (45)	NOT NULL	ID value
verificationStatus	VARCHAR (15)	DEFAULT 'Pending' ENUM ('Pending', 'Verified', 'Rejected')	Verification status
documentURL	VARCHAR (500)	NULLABLE	Object path within the private government-ids Storage bucket

3.23 TASKS

Stores volunteer tasks created by staff for a specific shelter branch.

Column Name	Data Type	Constraints	Description
taskID	INT	PK, AUTO_INCREMENT	Unique identifier for the task
taskName	VARCHAR (100)	NOT NULL ENUM ('Animal_Care', 'Vet_Assistance', 'Cleaning', 'Feeding', 'Events', 'Admin', 'Other')	Task Name
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter where task is assigned
taskDesc	VARCHAR (300)	NOT NULL	Task Description
taskDate	DATETIME	NULLABLE	Task creation date
taskDue	DATETIME	NULLABLE	Task due date
taskStatus	VARCHAR (45)	NULLABLE ENUM ('In_progress', 'Completed', 'Cancelled')	Task status
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff responsible for task

3.24 VOLUNTEER_TASK

Junction table linking volunteers to the tasks assigned to them.

Column Name	Data Type	Constraints	Description
taskID	INT	PK, FK → TASKS (taskID)	Unique identifier for the task
volunteerID	INT	PK, FK → VOLUNTEER (userID)	Unique identifier for the volunteer assigned to task

3.25 EVENTS

Stores adoption events, fundraisers and community programs organised by shelter staff.

Column Name	Data Type	Constraints	Description
eventID	INT	PK, AUTO_INCREMENT	Unique identifier for the event
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter where event takes place.

Column Name	Data Type	Constraints	Description
eventName	VARCHAR (45)	NOT NULL	Name of event
eventCategory	VARCHAR (25)	NOT NULL ENUM ('Adoption_Event', 'Fundraiser', 'Volunteer_Orientation', 'Vaccination_Clinic', 'Community_Outreach', 'Workshop', 'Donation_Drive', 'Other')	Event category
eventDate	DATETIME	NOT NULL	Event date and time
eventDesc	VARCHAR (300)	NOT NULL	Description of event.
staffID	INT	FK → STAFF (userID) NULLABLE	Unique identifier for the staff responsible for managing event

3.26 VOLUNTEER_EVENT

Junction table linking volunteers to the events they are participating in.

Column Name	Data Type	Constraints	Description
eventID	INT	PK, FK → EVENTS (eventID)	Unique identifier for the event
volunteerID	INT	PK, FK → VOLUNTEER (userID)	Unique identifier for the volunteer assigned to event

3.27 DONOR

Stores account and contact information for users who make monetary donations to shelter branches.

Column Name	Data Type	Constraints	Description
userID	INT	PK, FK → USERS (userID)	Unique identifier for the donor, shared with USERS
avatarSeed	VARCHAR (64)	NOT NULL	Seed string used to generate the user's DiceBear avatar
donorName	VARCHAR (45)	NOT NULL	Name of donor
addressLine1	VARCHAR (100)	DEFAULT ""	Street address, line 1
addressLine2	VARCHAR (100)	NULLABLE	Street address, line 2
city	VARCHAR (45)	DEFAULT ""	City

Column Name	Data Type	Constraints	Description
state	VARCHAR (45)	DEFAULT ''	State
zip	VARCHAR (10)	DEFAULT ''	Postal code
country	VARCHAR (45)	DEFAULT ''	Country
onboardingComplete	BOOLEAN	DEFAULT FALSE	Whether the donor has finished the onboarding wizard
onboardingStep	INT	DEFAULT 2	Furthest onboarding step reached (only ever advances)
donorPhone	VARCHAR (20)	NULLABLE	Donor contact number
donorDOB	DATE	NULLABLE	Date of Birth of donor
donorSex	CHAR (1)	NULLABLE	Gender of donor
createdAt	DATETIME	NOT NULL	Account creation timestamp
stripeCustomerID	VARCHAR (255)	NULLABLE	Stripe payment ID
accountStatus	VARCHAR (45)	DEFAULT 'Active' ENUM ('Active', 'Banned', 'Deactivated')	Account status

3.28 DONATION

Stores individual donation transactions made by donors to specific shelter branches.

Column Name	Data Type	Constraints	Description
donationID	INT	PK, AUTO_INCREMENT	Unique identifier for the donation
donationCode	TEXT	NULLABLE, UNIQUE, GENERATED	Human-facing reference (e.g. "DON-00012") GENERATED ALWAYS STORED from donationID, read-only
donorID	INT	FK → DONOR (userID)	Unique identifier for the donor
shelterID	INT	FK → SHELTER (shelterID)	Unique identifier for the shelter
donationDate	DATETIME	NOT NULL DEFAULT now()	Donation date and time

Column Name	Data Type	Constraints	Description
donationAmt	FLOAT	NOT NULL	Amount donated.
donationDesc	VARCHAR (300)	NULLABLE	Description of donation

3.29 Constraints & Indexes

Lists the database-level constraints and indexes that sit outside individual column definitions, including partial, expression and spatial indexes.

Table	Constraint	Purpose
ADOPTION_APPLICATION	UNIQUE (adopterID, petID) WHERE applicationStatus IN ('Pending', 'Accepted')	One active application per adopter + pet
APPOINTMENTS	UNIQUE (petID, vetID, appointmentDate) WHERE appointmentStatus = 'Scheduled'	Blocks double-submitted bookings
GOVERNMENT_ID	UNIQUE (userID, userType)	One ID per user
SPECIES	UNIQUE (LOWER(speciesName))	Case-insensitive unique names
BREED	UNIQUE (speciesID, LOWER(breedName))	Case-insensitive unique within a species
SHELTER	GIST index on shelterLocation	Fast PostGIS proximity search

Note: Partial/expression indexes and generated columns can't be expressed in the Prisma schema, so they live in migration SQL (the baseline migration builds the full schema).

4. ER-Diagram

ER-Diagram: <https://dbdiagram.io/d/PetPals-69bf5ca6fb2db18e3bd4b303>